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EX NO: 8A

SIMULATION AND ANALYSIS OF BPSK MODULATION USING CONSTELLATION DIAGRAMS IN MATLAB

OBJECTIVES

i) To generate BPSK symbols from a binary data sequence using phase-shift keying principles.

ii) To plot and analyze the BPSK signal constellation in the In-Phase (I) and Quadrature (Q) plane.

iii) To investigate the effect of noise on constellation points and evaluate the resulting degradation in symbol detection performance.

PROGRAM (MATLAB CODE)

```
clc;

clear;

close all;

%% Objective 1: Generate BPSK symbols

N = 1000;

data = randi([0 1], N, 1);

% BPSK Mapping

bpsk_symbols = 2*data - 1;

%% Objective 2: Plot BPSK Signal Constellation

figure;

subplot(1,2,1);

scatter(real(bpsk_symbols), imag(bpsk_symbols), 30, 'filled');

hold on;

xline(0,'k','LineWidth',1.5);

yline(0,'k','LineWidth',1.5);

grid on;

axis equal;
```

```
xlim([-1.5 1.5]);
ylim([-1 1]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title('Ideal BPSK Constellation');
%% Objective 3: Add Noise and Evaluate Performance
SNR = 5;
rx_signal = awgn(bpsk_symbols, SNR, 'measured');
subplot(1,2,2);
scatter(real(rx_signal), imag(rx_signal), 15, 'filled');
hold on;
xline(0,'k','LineWidth',1.5);
yline(0,'k','LineWidth',1.5);
grid on;
axis equal;
xlim([-2.5 2.5]);
ylim([-2 2]);
xlabel('In-Phase (I)');
ylabel('Quadrature (Q)');
title(['Noisy BPSK Constellation (SNR = ', num2str(SNR), ' dB)']);
%% Symbol Detection
detected_bits = real(rx_signal) > 0;
% Calculate Errors
num_errors = sum(data ~= detected_bits);
BER = num_errors/N;
fprintf('Number of Transmitted Bits : %d\n', N);
fprintf('Number of Bit Errors      : %d\n', num_errors);
fprintf('Bit Error Rate (BER)      : %f\n', BER);
```

PROCEDURE

Objective (i): Generate BPSK Symbols

1. Initialize the MATLAB environment and define the number of bits.
2. Generate a random binary sequence consisting of 0s and 1s.
3. Apply BPSK modulation by mapping binary 0 to −1 and binary 1 to +1.
4. Store the generated BPSK symbols for further processing.
5. Verify that the symbol sequence contains only two distinct values.
6. Prepare the modulated symbols for constellation analysis.

Result:
BPSK symbols were successfully generated from the binary data sequence using phase-shift keying modulation.

Objective (ii): Plot and Analyze Constellation

1. Use the generated BPSK symbols for plotting.
2. Assign the real part to the In-Phase (I) axis.
3. Set the Quadrature (Q) component to zero.
4. Plot the constellation points on the I–Q plane using a scatter plot.
5. Draw horizontal and vertical axes through the origin.
6. Analyze symbol locations corresponding to binary 0 and 1.

Result:
The BPSK constellation was successfully plotted in the In-Phase (I) and Quadrature (Q) plane, showing two distinct symbol points corresponding to binary 0 and binary 1 for the given SNR.

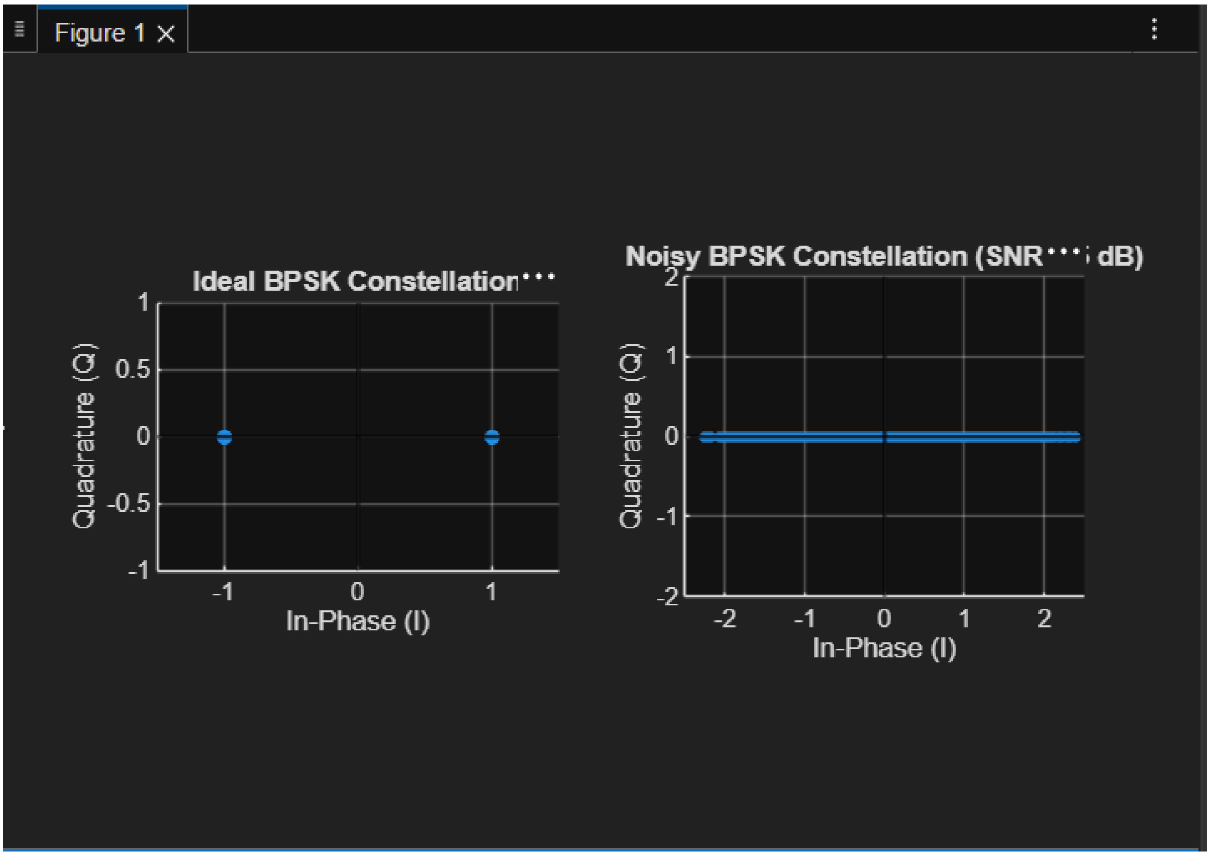
Objective (iii): Noise Effect and BER Analysis

1. Specify the Signal-to-Noise Ratio (SNR).
2. Add Additive White Gaussian Noise (AWGN) to the BPSK symbols.
3. Plot the noisy constellation diagram.
4. Observe dispersion around ideal symbol positions.
5. Detect symbols using a threshold decision rule.
6. Compare transmitted and received bits to compute BER.

Result:

The effect of AWGN noise on the constellation points was successfully analyzed, and the degradation in symbol detection performance was evaluated using the Bit Error Rate (BER).

OUTPUT :



FINAL RESULT :

The BPSK modulation was successfully simulated and analyzed using MATLAB. The constellation diagrams clearly showed ideal and noisy signal behavior. The addition of noise caused dispersion of constellation points, resulting in bit errors, which were quantified using BER.